Fra SOAR til Agentic SOC:
En GenAI-aktivert AI-verktøykasse for Microsoft Sentinel

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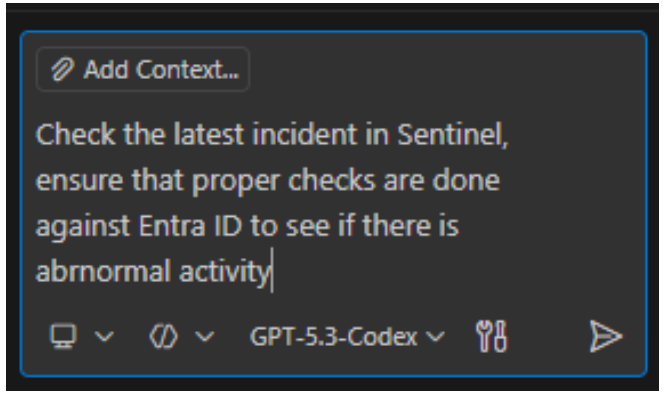
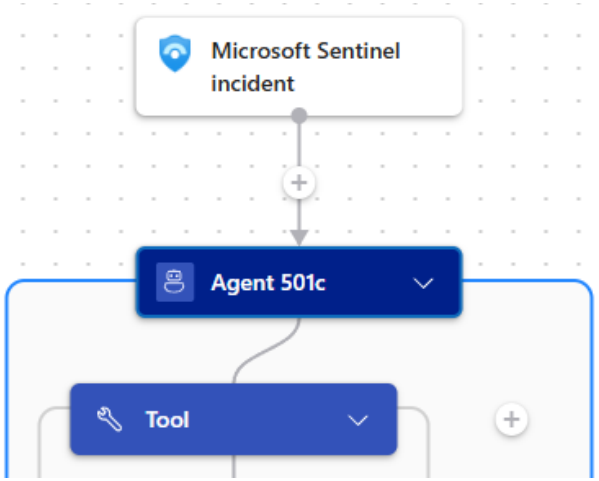
Microsoft® MVP Most Valuable Professional

Agentic SOC





Analytikeren

Agentic Workbench

IDE + MCP + CLI

Automatisering

Agentic SOAR

Runbook + LLM

C2 - Restricted use

GenAI Utviklingen fra 2022 - 2025

99,9% kostnadsreduksjon

90% Reduksjon i Hallusinasjoner (m/RAG)

Kan behandle **500 ganger** mer data

Opptil 1,5 time med video

15 ganger raskere

10.000+ Modeller klare til bruk

Kan kjøre overalt **(Mobil/PC/Sky)**

Nye standarder for kommunikasjon **(MCP/A2A)**

De Store Aktørene
OpenAI, Google, Anthropic, xAI og Meta

Underdogs
Mistral, Qwen, Deepseek

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Språkmodeller (Mars 2026)

- Best språkmodell til **utvikling** = **GPT 5.4 Codex xhigh**
- Best språkmodell til **agenter** = **Claude Opus 4.6**
- Best språkmodell til **multimodalitet** = **Gemini 3.1 Pro Preview**
- Best språkmodell til **reasoning** = **Gemini 3.1 Pro Preview**
- **Beste for sikkerhetsanalyse/logg/agenter = Claude Opus 4.6**
- Mest **treffsikker** språkmodell? = **GPT 5.4 Codex xhigh**
- Beste modellen til Computer-Use? = **GPT 5.4 Codex xhigh**
- Hvilken språkmodell kan håndtere mest **data** = **Grok 4.1 Fast**
- Beste språkmodellen som kan kjøre på egen infrastruktur? = **Kimi K2.5**
- Analyse av Knut Hamsun dikt? = **NorLLM**

Kontekst Vindu =
4 floppy disker

NB: Sist oppdatert 9 Mars kl 07:45.....

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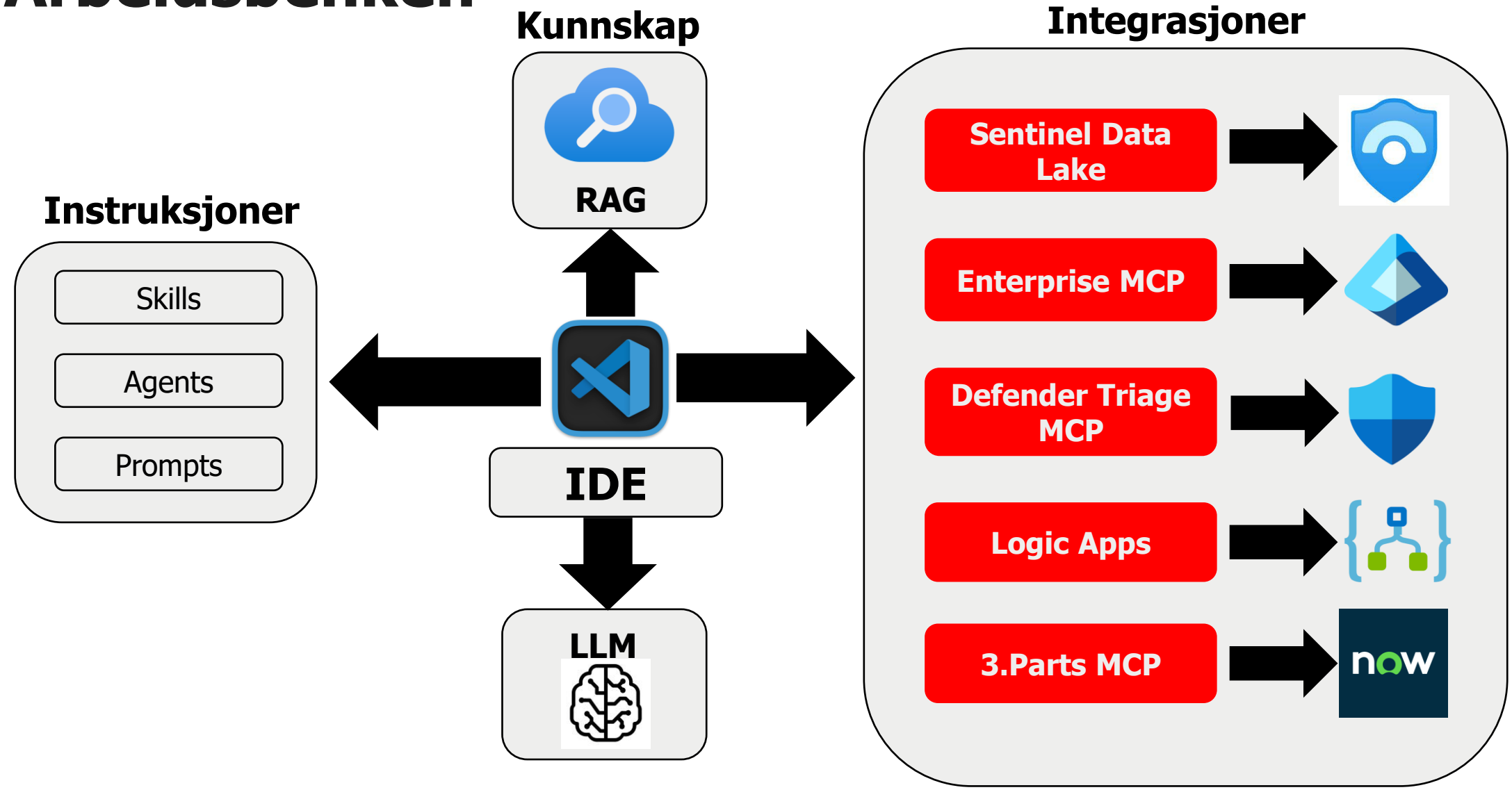
Arbeidsbenken



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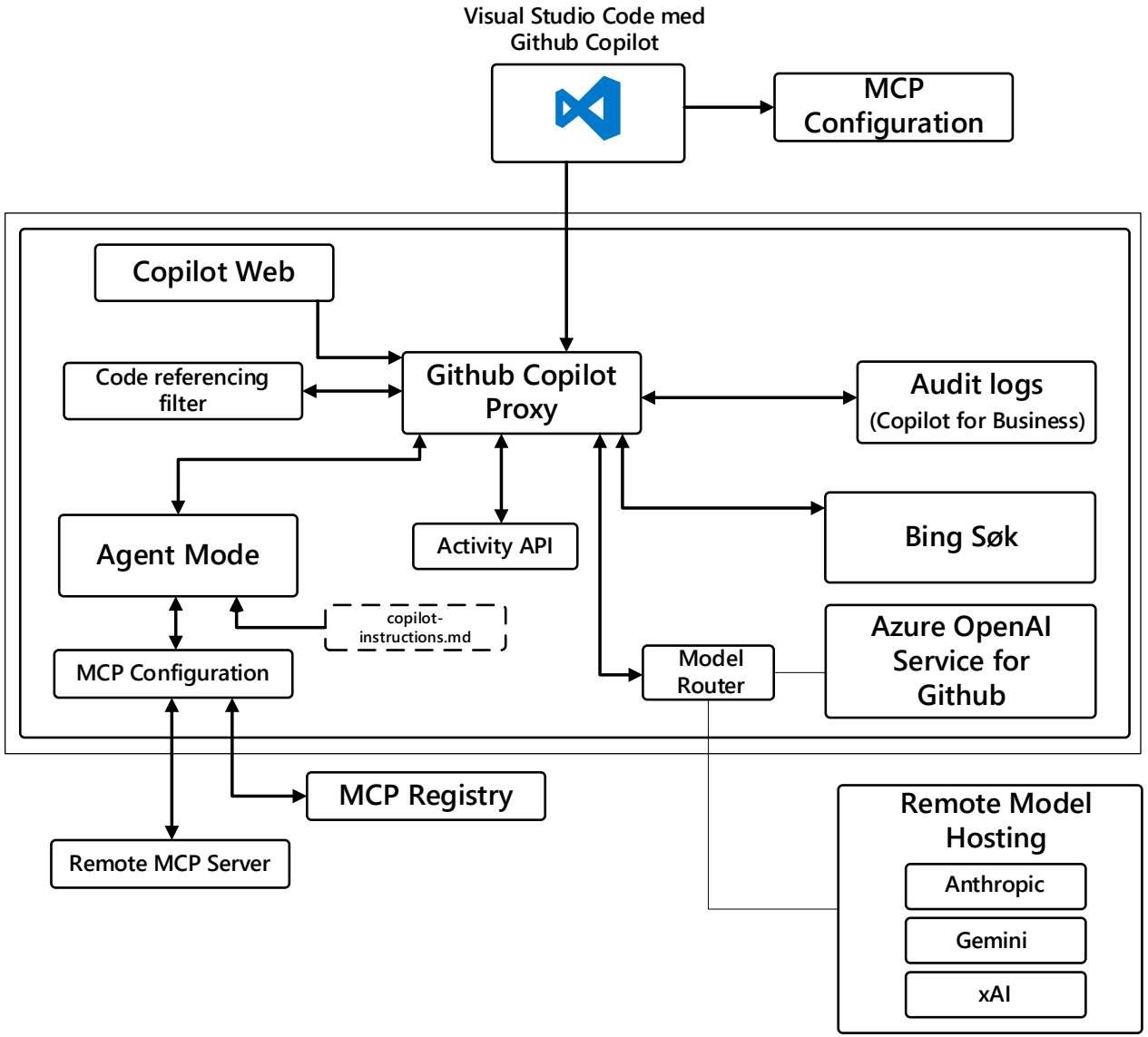
Github Copilot



Har begrensninger på antall tokens per LLM (maks 272k tokens)

LLMer godkjennes via GitHub org config

Agenter kan kjøre via Cloud runtime

Github Agenter kan også bruke f.eks MCP mot Sentinel eller egenutviklet

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VS Code Struktur

.github**prompts** (Inneholder gjenbrukbare maler for predefinerte spørringer)

.github**skills** (Definerer spesifikke ferdigheter som AI-agentene kan bruke)

.github**agents** (Inneholder konfigurasjonsfiler for spesialiserte AI-agenter)

.github**copilot-instructions.md** (overordnede regler og instruksjoner for Copilot)

.vscode**mcp.json** (MCP Konfig for VS Code)

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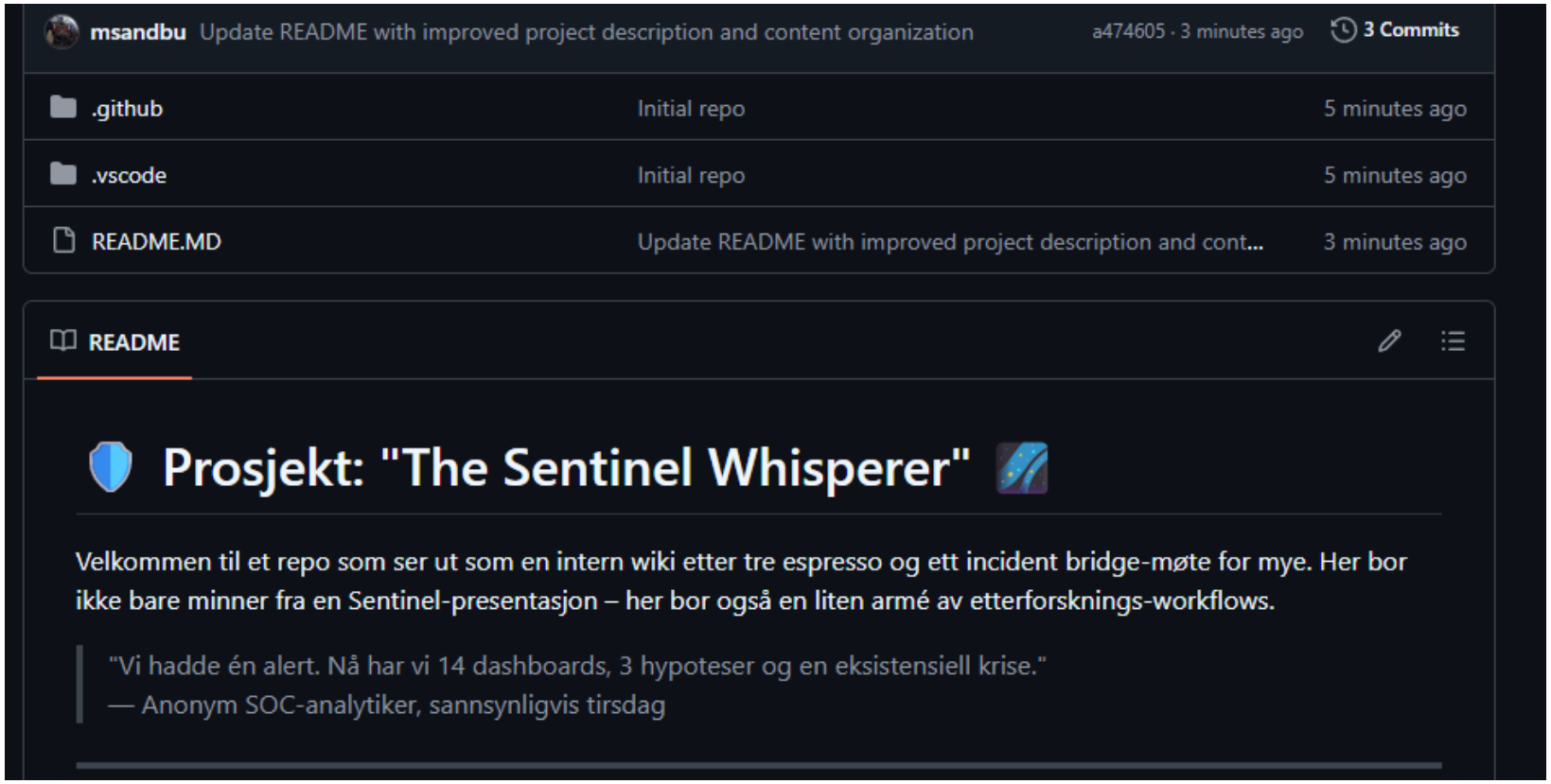
Ferdig eksempel



https://github.com/msandbu/sentinelday

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Oppsett

Prompt →

"Check for the latest incidents in Sentinel and determine what kind of IP addresses are used and check them against AbuseDB"

MCP Config

Skills

Copilot-intructions

SENTINEL INCIDENT REVIEW AND IP ADDRESS CHECK

Latest Sentinel incidents

- Most recent incidents are RDP brute-force themed; top two are still `New`: `#36` and `#35` (both Medium).
- In the latest 10 incidents, 5 include IPv4 entities (`#14`, `#15`, `#12`, `#2`, `#11`), all tied to `Failed logon attempt RDP (Bruteforce)` (High, Closed).

IP type determination

- Extracted 10 unique source IPv4s from incident-linked alert entities.
- All 10 are **public IPv4** (no private/RFC1918, loopback, or link-local IPs detected).

AbuseIPDB (AbuseDB) check

- 🔴 Very high abuse confidence (100):
  `109[.]205[.]211[.]14` (419 reports), `185[.]156[.]73[.]173` (116), `185[.]156[.]73[.]59` (111), `185[.]156[.]73[.]69` (133), `185[.]156[.]73[.]24` (95), `92[.]63[.]197[.]69` (89), `92[.]63[.]197[.]9` (113)
- 🟠 Medium abuse confidence:
  `185[.]156[.]73[.]74` (54, 36 reports), `185[.]156[.]73[.]169` (51, 29 reports)

Todos (5/5)

Add Context...

Describe what to build next

Local Agent GPT-5.3-Codex

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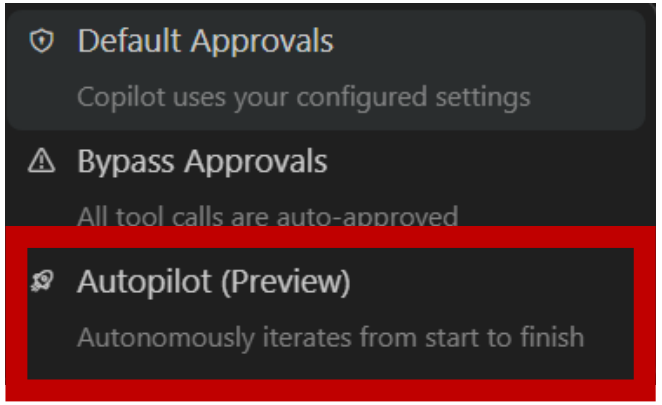
Agents / Skills / Prompt

Agent.md = Kontekst + Modell + Verktøy + Subagenter (Isolert)

```
---
description: "Use when summarizing Microsoft Sentinel incidents for
executive leadership, CISO briefings, board-ready security summaries, top
10 risks, priority focus areas, incident trend highlights, and strategic
response recommendations."
name: "Sentinel CISO Incident Summarizer"
argument-hint: "Describe the time range, scope (workspace/team/business
unit), and whether you want daily, weekly, or monthly top 10 priorities."
user-invocable: true
Configure Tools...
tools: [execute, agent]
```



Skill.md = Kontekst + Modell + Verktøy

```
---
name: computer-investigation
description: Use this skill when asked to investigate a computer, device, endpoint, or machine for security issues, suspicious activity,
malware, or compliance review. Triggers on keywords like "investigate computer", "investigate device", "investigate endpoint", "check
machine", "device security", "endpoint investigation", or when a device name/hostname is mentioned with investigation context. This skill
provides comprehensive device security analysis including Defender alerts, sign-in patterns, logged-on users, vulnerabilities, software
inventory, compliance status, network activity, and automated investigation tracking for Entra Joined, Hybrid Joined, and Entra Registered
devices.
```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RAG (Retrieval Augmented Generation)

- **Søkefunksjon**
  - **Eks:** ITSM, Microsoft 365, Jira eller AI Search
  - Bruk **WorkIQ MCP** for Microsoft 365 kilder
  - AI Search støtter multimodalitet (tekst & bilde)
- **Brukes via MCP eller Agent**
- **Finne relevant data for LLM**
  - ✓ Kunnskapsartikler
  - ✓ Tidligere løste saker
  - ✓ Prosesser og rutiner

Funn

- Hoveddokument: `bruteforce.txt` med prosess for "Handling Brute Force Attacks on RDP in Microsoft Azure".
- **Containment**: slå av offentlig RDP, begrens RDP med `NSG` + `JIT VM Access`, tillat kun fra betrodde IP-er/VPN, og lås/deaktiver kontoer med gjentatte feilforsøk.
- **Forebygging**: bruk `Azure Bastion` i stedet for direkte RDP, krev `VPN/Private Endpoint`, håndhev `MFA`, sterke passord, aktiver `Defender for Servers`, og bruk Sentinel-regler for brute force-deteksjon.

Eks med Azure AI Search

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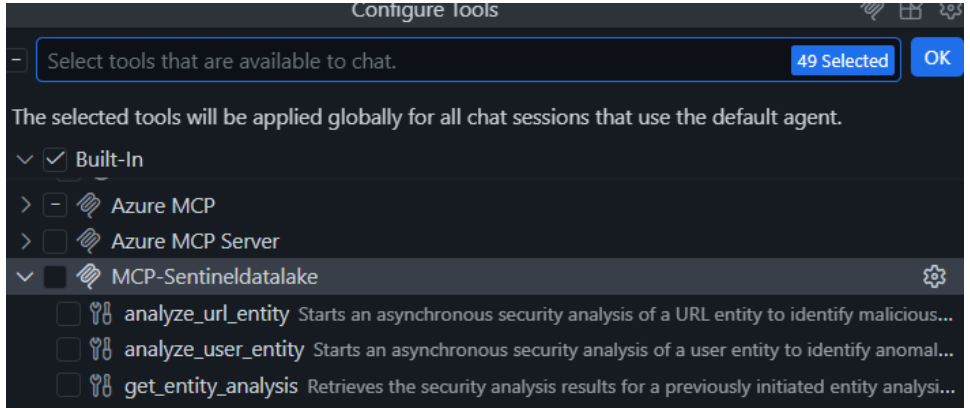
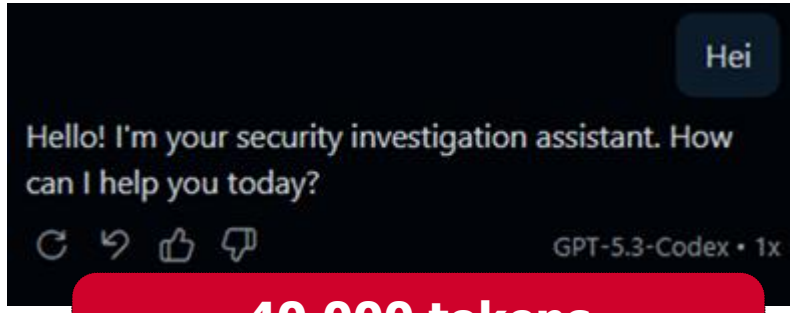
Ting man bør være klar over

- **MCP Autentisering via Oauth**
- **Bør ikke ha flere enn 128 tools (MCP)**
  - Styres enten sentralt via MCP Gateway eller individuelt
  - MCP Gateway = API Management / Context-Forge
  - [Optimizing Developer Experience with AI and MCP - msandbu.org]
- **Sentralt repo med Codespaces**
- **MCP Spiser tokens**
- **Agent.md > Skills.md – Høyere treffsikkerhet**
- **MCP Server for Data Lake**
  - Pass på |TimeGenerated legges ved I prompten
  - Kan fort generere masse kostander





40,000 tokens med 5 MCP Servere

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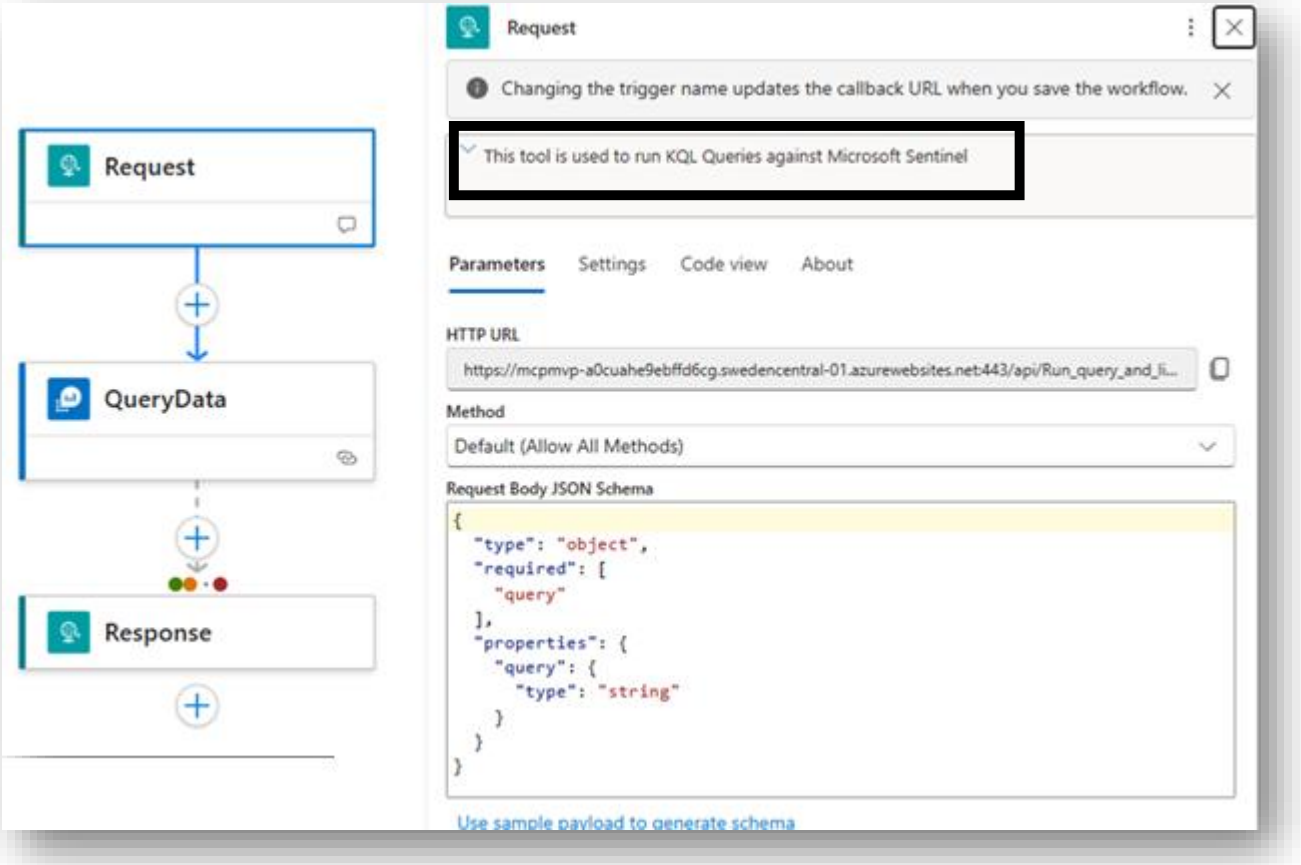
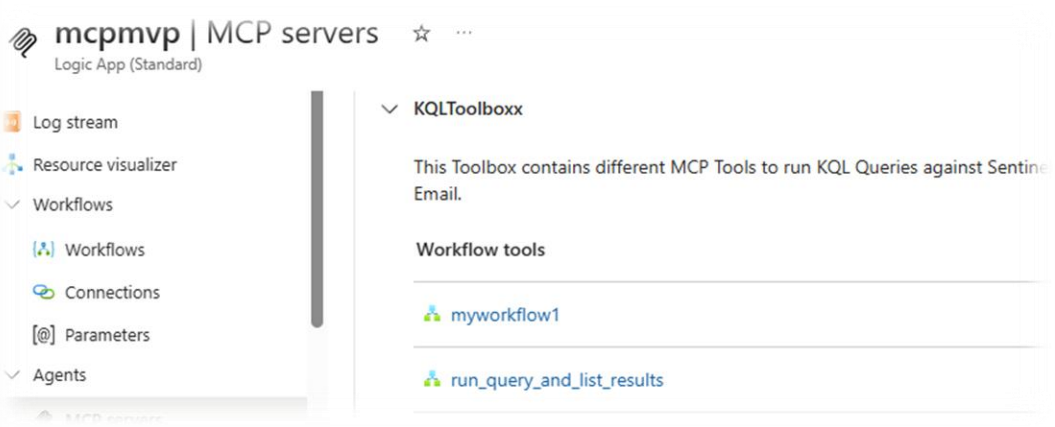
Egenutviklet MCP servere med Logic App





VS Code Konfigurasjon (mcp.json)

```
},
"Sentineltoolbox": {
"url": "https://logicapp/api/mcpservers/navn/mcp",
"type": "http"
},
```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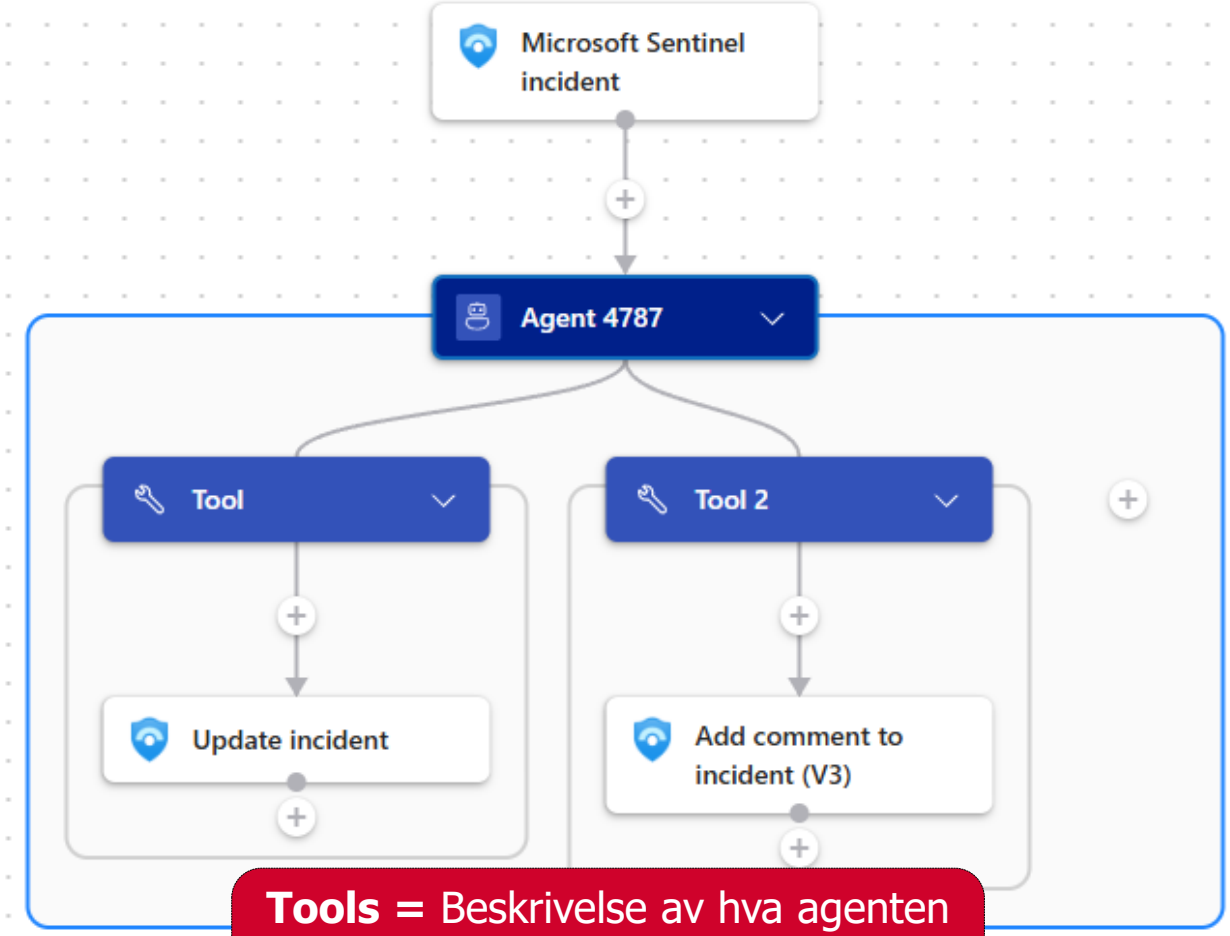
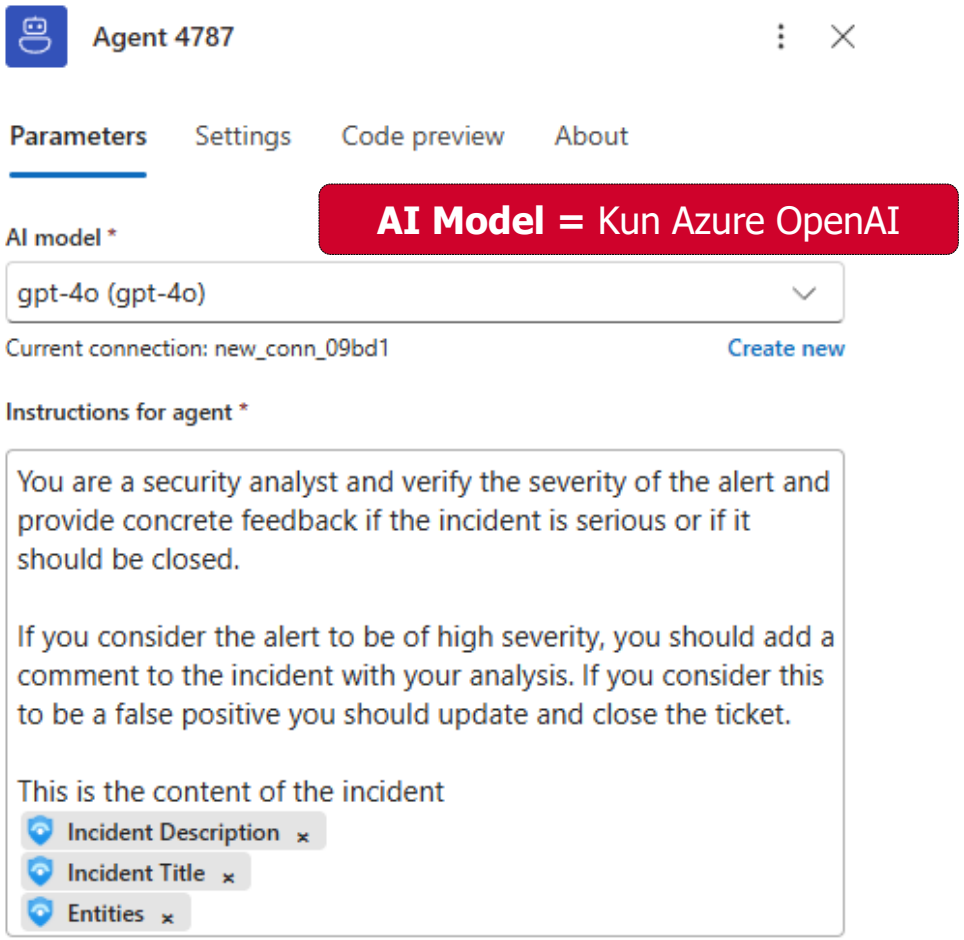
Automatisering med Agentic Loop



Tools = Beskrivelse av hva agenten skal gjøre av oppgaver basert på informasjonen den får



AI Model = Kun Azure OpenAI

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Demo

1. Arbeidsbenk med Visual Studio Code + MCP + Skills + Agenter
2. Utvikling av MCP Server via Logic App
3. Automatisering med Playbooks og Agentic Loops

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Oppsummering

Linker til MCP Servere og inspirasjon

- https://github.com/markolauren/ResponseMCP
- https://github.com/stefanpems/ai-powered-soc
- https://github.com/SCStelz/security-investigator
- https://github.com/MHaggis/Security-Detections-MCP
- **LLM Benchmarks:** https://artificialanalysis.ai/

https://github.com/msandbu/sentinelday

Spørsmål? Noe du lurer på?

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